

Applied PHYSICS

Assignment # 03

CHAPTER # 21

Problem # 3

Given Data:

$$\text{Force} = 5.70 \text{ N}$$

$$q_1 = 26.0 \text{ MC} = 26 \times 10^{-6} \text{ C}$$

$$q_2 = -47.0 \text{ MC} = -47 \times 10^{-6} \text{ C}$$

To Find:

$$\text{Distance} = r = ?$$

Solution:

$$\text{As } F = \frac{k q_1 q_2}{r^2}$$

$$r^2 = \frac{k q_1 q_2}{F}$$

Taking square root on b/s

$$r = \sqrt{\frac{k q_1 q_2}{F}}$$

As we know

$$\text{So, } k = 8.99 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$$

$$r = \sqrt{\frac{(8.99 \times 10^9) (26 \times 10^{-6}) (-47 \times 10^{-6})}{5.70}}$$

$$r = 1.39 \text{ m}$$

Result:

The distance between q_1 and q_2 is 1.39m

QUESTION#5

Given Data:

$$q_1 = 3 \times 10^{-6} \text{ C}$$

$$q_2 = 1.50 \times 10^{-6} \text{ C}$$

$$r = 12 \text{ cm} = \frac{12}{100} = 0.12 \text{ m}$$

$$k = 8.99 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$$

To Find:

Magnitude Force = $F = ?$

Solution:

As

$$F_E = \frac{k q_1 q_2}{r^2}$$

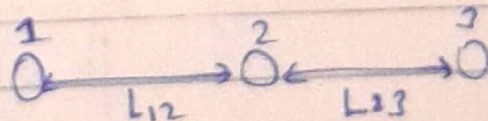
$$F_E = \frac{(8.99 \times 10^9) (3 \times 10^{-6}) (1.5 \times 10^{-6})}{(0.12)^2}$$

$$F_E = 2.81 \text{ N}$$

Result:

The magnitude of Electrostatic Force between particles is 2.81 N.

QUESTION#7



Given Data

$$L_{23} = L_{12}$$

To Find:

$$\text{Ratio} = \frac{q_1}{q_2} = ?$$

Solution:

$$F_{\text{net}} = F_{13} + F_{23} \quad \text{--- (1)}$$

$$r_{13} = L_{12} + L_{23} \\ = L_{23} + L_{23} \quad \because L_{12} = L_{23} \text{ (Given)}$$

$$r_{13} = 2L_{23}$$

So, (1) become

$$F_{\text{net}} = \frac{kq_1q_3}{(2L_{23})^2} + \frac{kq_2q_3}{(L_{23})^2}$$

$$0 = \frac{kq_1q_3}{4(L_{23})^2} + \frac{kq_2q_3}{(L_{23})^2}$$

$$\frac{kq_1q_3}{4(L_{23})^2} = - \frac{kq_2q_3}{(L_{23})^2}$$

$$\frac{q_1}{4} = -q_2$$

$$\boxed{\frac{q_1}{q_2} = -4}$$

Result:

The ratio $\frac{q_1}{q_2}$ is -4

QUESTION #10

Given Data:

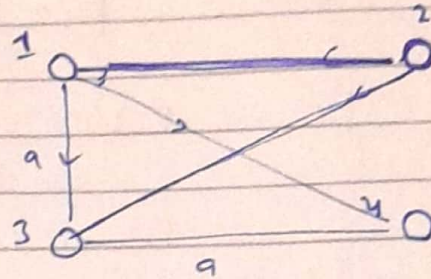
$$q_1 = q_4 = Q$$

$$q_2 = q_3 = q$$

To Find:

$$\text{ratio} = \frac{Q}{q} = ?$$

Solution:



(a)

Find net

At Particle 1

→ we consider along y-axis

$$\begin{aligned} H^2 &= B^2 + P^2 \\ H &= \sqrt{a^2 + a^2} \\ r &= \sqrt{2a^2} = \sqrt{2}a \end{aligned}$$

$$F_{\text{net } y} = F_{31} + F_{41} \sin 45^\circ$$

$$0 = \frac{kqQ}{a^2} + \frac{kQq}{(\sqrt{2}a)^2} \sin 45^\circ \quad \because q_1 = q_4 = Q$$

$$\because q_2 = q_3 = q$$

$$\frac{-kqQ}{a^2} = \frac{kQq}{2a^2} \sin 45^\circ$$

$$-q = \frac{Q}{2} \cdot \frac{1}{\sqrt{2}}$$

$$-2\sqrt{2} = \frac{Q}{q}$$

$$\boxed{\frac{Q}{q} = -2.8284}$$

(b)

At particle 2

→ we consider along x-axis

$$F_{\text{net } x} = F_{21} + F_{23} \cos 45^\circ$$

$$0 = \frac{kqQ}{a^2} + \frac{kqQ}{(\sqrt{2}a)^2} \cos 45^\circ$$

$$-\frac{kqQ}{a^2} = \frac{kqQ}{2a^2} \cdot \frac{1}{\sqrt{2}}$$

$$-Q = \frac{q}{2} \cdot \frac{1}{\sqrt{2}}$$

$$\frac{Q}{q} = -\frac{1}{2\sqrt{2}}$$

$$\frac{Q}{q} = -0.3535$$

Result:

(a) The ratio of $\frac{Q}{q}$ at particle 1 is $-2\sqrt{2}$

(b) The ratio of $\frac{Q}{q}$ at particle 2 is $-\frac{1}{2\sqrt{2}}$

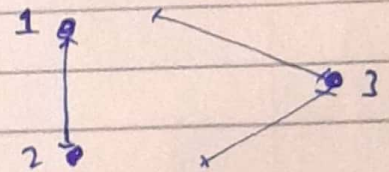
QUESTION # 17

Given Data:

$$q = q_1 = q_2 = 20 \mu\text{C} = 20 \times 10^{-6} \text{C}$$

$$d = 1.50 \text{m}$$

$$k = 8.99 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$



To Find:

(a) Force on particle 1 due to 2 = $F_{12} = ?$

(b) Force on particle 1 due to 2 and 3 = F_y

Solution:

(a)

$$F_{12} = \frac{k q_1 q_2}{d^2}$$

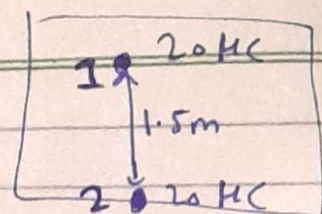
$$F_{12} = \frac{k q q}{d^2}$$

$$\therefore q_1 = q_2 = q$$

$$F_{12} = \frac{k q^2}{d^2}$$

$$= \frac{(8.99 \times 10^9) (20 \times 10^{-6})^2}{(1.5)^2}$$

$$\boxed{F_{12} = 1.60 \text{ N}}$$



(b)

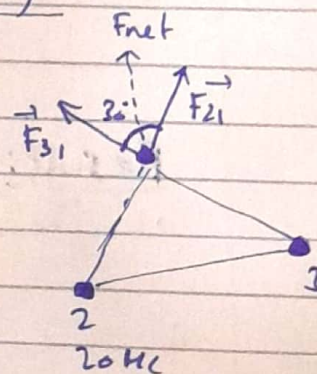
Magnitude of F_{net}

$$F_y = 2 \left(\frac{k q^2}{d^2} \right) \cos 30^\circ$$

$$F_y = 2 \left(\frac{8.99 \times 10^9 \times (20 \times 10^{-6})^2}{(1.5)^2} \right) \cos 30^\circ$$

$$= 2 \left(\frac{8.99 \times 10^9 \times (20 \times 10^{-6})^2}{(1.5)^2} \right) \cdot \frac{\sqrt{3}}{2}$$

$$\boxed{F_y = 2.77 \text{ N}}$$



Result:

(a) The Force on particle 1 due to particle 2 is 1.60 N

(b) The Force on particle 1 due to particle 2 and 3 is 2.77 N

QUESTION # 24

Given Data:

$$\text{Identical charge} = q = -1 \times 10^{-16} \text{ C}$$

$$q_1 = q_2 = q$$

$$\text{Distance} = r = 1 \text{ cm} = 1 \times 10^{-2} \text{ m}$$

$$k = 8.99 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$$

To Find:

(a) Force = $F = ?$

(b) No. of Electron = $n = ?$

Solution:

(a) $F = \frac{kq_1q_2}{r^2}$

$$F = \frac{kq^2}{r^2}$$

$$F = \frac{(8.99 \times 10^9) (-1 \times 10^{-16})^2}{(1 \times 10^{-2})^2}$$

$$\boxed{F = 8.99 \times 10^{-19} \text{ N}}$$

$$\text{No. of electron} = n = \frac{q}{e}$$

(b)

$$\text{No. of electron} = n = \frac{q}{e}$$

$$\text{As } e = 1.6 \times 10^{-19}$$

$$n = \frac{-1 \times 10^{-16}}{1.6 \times 10^{-19}} = 625$$

Result:

- (a) The magnitude of Electrostatic Force acting between the particle is $8.99 \times 10^{-19} \text{ N}$
- (b) Number of electron are 625.

QUESTION # 25

Given Data:

$$\text{Charge} = q = 1 \times 10^{-7} \text{ C}$$
$$e = 1.6 \times 10^{-19}$$

To Find:

$$\text{No. of electron} = n = ?$$

Solution:

$$n = \frac{q}{e}$$

$$= \frac{1 \times 10^{-7}}{1.6 \times 10^{-19}}$$

$$\boxed{n = 6.3 \times 10^{11}}$$

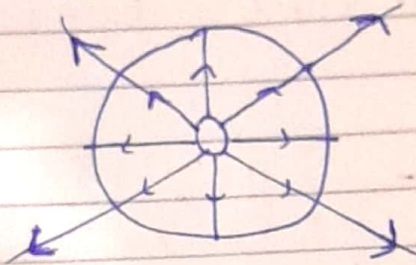
Result:

The electron that have to be removed are 6.3×10^{11} .

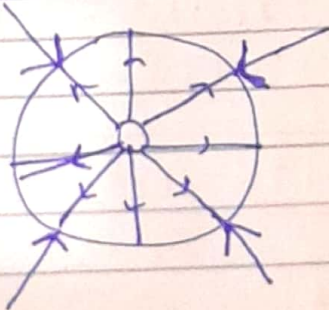
CHAPTER # 22

QUESTION #1 sketch Electric Field lines between two concentric conducting spherical shells. When uniform positive charge is inner and uniform -ve charge $-q_2$ is outer.

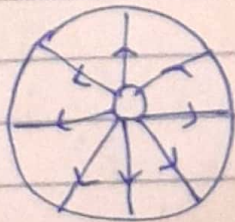
For $q_1 > q_2$



For $q_1 < q_2$



For $q_1 = q_2$



QUESTION #4

Given Data:

$$\begin{aligned} q_1 &= -2 \times 10^{-7} \text{ C} & \text{at } x &= 13.5 \text{ cm} = 13.5/100 = 0.135 \text{ m} \\ q_2 &= 2 \times 10^{-7} \text{ C} & \text{at } x_1 &= 6 \text{ cm} = 0.06 \text{ m} \\ & & \text{at } x_2 &= 21 \text{ cm} = 0.21 \text{ m} \end{aligned}$$

To Find:

$$\text{Electric Field} = \vec{E}_1 = ?$$

$$E_2 = ?$$

$$\text{Net Field} = \vec{E}_1 + \vec{E}_2 = ?$$

Solution:

$$\begin{aligned} \vec{E}_1 &= - \frac{q_1}{4\pi\epsilon_0 (x - x_1)^2} \hat{i} \\ \vec{E}_1 &= - \frac{k q_1}{(x - x_1)^2} \hat{i} \\ &= - \frac{(8.99 \times 10^9) (-2 \times 10^{-7})}{(0.135 - 0.06)^2} \hat{i} \end{aligned}$$

$$\boxed{\vec{E}_1 = - (3.196 \times 10^5 \text{ N C}^{-1}) \hat{i}}$$

$$\begin{aligned} \vec{E}_2 &= - \frac{k q_2}{(x - x_2)^2} \hat{i} \\ &= - \frac{(8.99 \times 10^9) (2 \times 10^{-7})}{(0.135 - 0.21)^2} \hat{i} \end{aligned}$$

$$\text{Thus, } \vec{E}_2 = - (3.196 \times 10^5 \text{ N C}^{-1}) \hat{i}$$

$$\text{Net Field} = \vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2$$

$$\begin{aligned} &= - 3.196 \times 10^5 - 3.196 \times 10^5 \\ \boxed{\vec{E}_{\text{net}} = - (6.39 \times 10^5 \text{ N C}^{-1}) \hat{i}} \end{aligned}$$

Result: The electric field $\vec{E}_1$ and $\vec{E}_2$ are $-3.196 \times 10^5 \text{ NC}^{-1}$ and $-3.196 \times 10^5 \text{ NC}^{-1}$ respectively.
Thus Net Field is $-6.39 \times 10^5 \text{ NC}^{-1}$

QUESTION # 05:

Electric Field = $E = 2 \text{ NC}^{-1}$

Distance = $r = 50 \text{ cm} = \frac{50}{100} = 0.5 \text{ m}$

To Find:

Magnitude of particle charge = $|q| = ?$

Solution:

As we know that

$$E = \frac{|q|}{4\pi\epsilon_0 r^2}$$

$$\therefore k = \frac{1}{4\pi\epsilon_0}$$

$$E = \frac{k|q|}{r^2}$$

$$\Rightarrow |q| = \frac{E r^2}{k}$$

$$|q| = E \cdot k r^2$$

$$= (2) \frac{(8.99 \times 10^9) (0.5)^2}{(8.99 \times 10^9)}$$

$$\boxed{|q| = 5.6 \times 10^{-11} \text{ C}}$$

Result:-

The magnitude of particle charge $5.6 \times 10^{-11} \text{ C}$.

QUESTION # 07

Given Data:

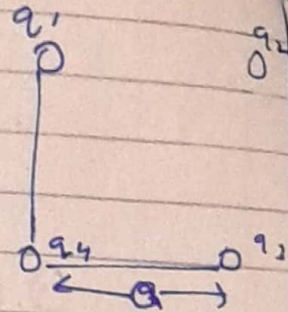
$$\text{Length} = a = 5 \text{ cm}$$

$$q_1 = 10 \text{ nC}$$

$$q_2 = -20 \text{ nC}$$

$$q_3 = +20 \text{ nC}$$

$$q_4 = -10 \text{ nC}$$



To Find:

$$\text{Net Electric Field} = ?$$

$$E_x + E_y = ?$$

$$r = \frac{a}{\sqrt{2}}$$

Solution:

Electric Field Along x-axis

$$E_{\text{net } x} = E_{1x} + E_{2x} + E_{3x} + E_{4x}$$
$$E_{\text{net } x} = k \left[\frac{|q_1|}{(a/\sqrt{2})^2} + \frac{|q_2|}{(a/\sqrt{2})^2} - \frac{|q_3|}{(a/\sqrt{2})^2} - \frac{|q_4|}{(a/\sqrt{2})^2} \right] \cdot \cos 45^\circ$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{a^2/2} \left[|q_1| + |q_2| - |q_3| - |q_4| \right]$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{a^2/2} \cdot \frac{1}{\sqrt{2}} \left[|q_1| + |q_2| - |q_3| - |q_4| \right]$$

Putting the values

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\sqrt{2}}{a^2} (10 + 20 - 20 - 10)$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\sqrt{2}}{a^2} (0)$$

$$\boxed{E_{\text{net } x} = 0}$$

$$E_{\text{net } y} = E_{1y} + E_{2y} + E_{3y} + E_{4y}$$

$$= -\frac{kq_1 \sin \theta}{\left(\frac{a}{\sqrt{2}}\right)^2} + \frac{kq_2 \sin \theta}{\left(\frac{a}{\sqrt{2}}\right)^2} + \frac{kq_3 \sin \theta}{\left(\frac{a}{\sqrt{2}}\right)^2} - \frac{kq_4 \sin \theta}{\left(\frac{a}{\sqrt{2}}\right)^2}$$

$$= \frac{-k}{\frac{a^2}{2}} \sin \theta [-|q_1| + |q_2| + |q_3| - |q_4|]$$

$$= \frac{-k}{\frac{a^2}{2}} \cdot \frac{1}{\sqrt{2}} [-|q_1| + |q_2| + |q_3| - |q_4|]$$

Putting values

$$= \frac{-k \sqrt{2}}{a^2} (-10 + 20 + 20 - 10)$$

$$= \frac{(8.99 \times 10^9) 2 \times 10^{-8} \sqrt{2}}{(0.05)^2}$$

$$E_y = 1.02 \times 10^5 \text{ NC}^{-1}$$

Thus,

$$E_{\text{net}} = E_{\text{net } x} + E_{\text{net } y}$$

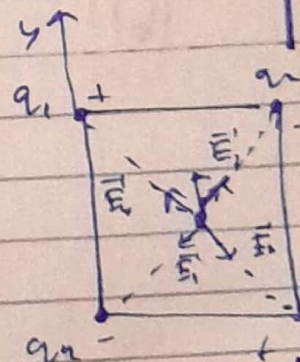
$$= 0 + 1.02 \times 10^5$$

$$E_{\text{net}} = 1.02 \times 10^5 \text{ NC}^{-1}$$

The Field pointing to +y direction is the vector sum of electric field of individual charges.

Result:

The net Electric Field is $1.02 \times 10^5 \text{ NC}^{-1}$



QUESTION # 9

Given Data:

$$\begin{aligned} q &= 3.20 \times 10^{-19} \text{ C at } x = +3 \text{ m} \\ -q &= -3.2 \times 10^{-19} \text{ C at } x = -3 \text{ m} \\ y &= 4 \text{ m}, d = 3 \text{ m} \\ k &= 8.99 \times 10^9 \text{ Nm}^2 \text{ C}^{-2} \end{aligned}$$

To Find:

Magnitude = $E_{\text{net } x} = ?$
Direction = ?

Solution:

(a)

$$\begin{aligned} E_{\text{net } x} &= \frac{2 |q| d}{4\pi\epsilon_0 (d^2 + y^2)^{3/2}} \\ &= \frac{2 (8.99 \times 10^9) (3.2 \times 10^{-19}) (3)}{[(3)^2 + (4)^2]^{3/2}} \end{aligned}$$

$$E_{\text{net } x} = 1.38 \times 10^{-10} \text{ N/C}$$

(b)

The net electric Field points in $-x$ direction
or 180° Counter Clockwise from $+x$ axis.

RESULT

- (a) The magnitude ($E_{\text{net } x}$) is $1.38 \times 10^{-10} \text{ N/C}$.
(b) The direction is along $-x$ direction.

QUESTION # 18

Given Data:

$$E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3} + E_{\text{next}} = ?$$

To Find:

$$E_{\text{next}} = ?$$

Calculation:

As we know that $E = \frac{1}{4\pi\epsilon_0} \frac{2qd}{z^4 \left(1 - \frac{d^2}{4z^2}\right)^2}$

$$E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3}$$

→ For finding next term we use binomial Expression.

$$E = \frac{q}{4\pi\epsilon_0 z^3} \left[\left(1 + \frac{d}{z} + \frac{3}{4} \frac{d^2}{z^2} + \frac{1}{2} \frac{d^3}{z^3} + \dots \right) - \left(1 - \frac{d}{z} + \frac{3}{4} \frac{d^2}{z^2} + \frac{1}{2} \frac{d^3}{z^3} \right) \right]$$

$$E = \frac{qd}{2\pi\epsilon_0 z^3} + \frac{qd^3}{4\pi\epsilon_0 z^5} + \dots$$

Therefore,

$$E_{\text{next}} = \frac{qd^3}{4\pi\epsilon_0 z^5}$$

Result: The next term for the electric field would be $\frac{qd^3}{4\pi\epsilon_0 z^5}$